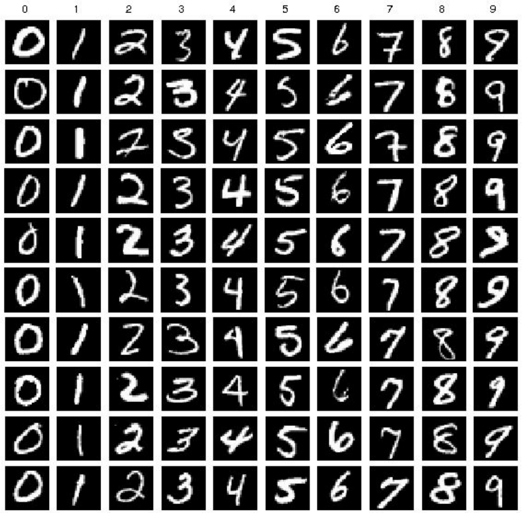
Generative Adversarial Network (GAN) application

What is GAN?

Basically, GAN can learn the pattern from the dataset and generate the fake images that are similar to the images in the dataset.



Dataset

Generated Image

GAN application

Basically, if we have large dataset, GAN could generate the world. Here is some interesting application of GANs.

- Human face
- Realistic photo
- Anime character
- Multiple features of face
- Face combination
- Text to image
- Image inpainting
- Image to image translation
- ....

Human face

Based on CeleA dataset (face of celebrities), below images are generated by GAN

Figure 5: 1024×1024 images generated using the CELEBA-HQ dataset. See Appendix F for a larger set of results, and the accompanying video for latent space interpolations.

Realistic photo

Yeah, it’s not real photo, generated by GAN

Anime character

GAN can auto generate and

colorize the Anime character

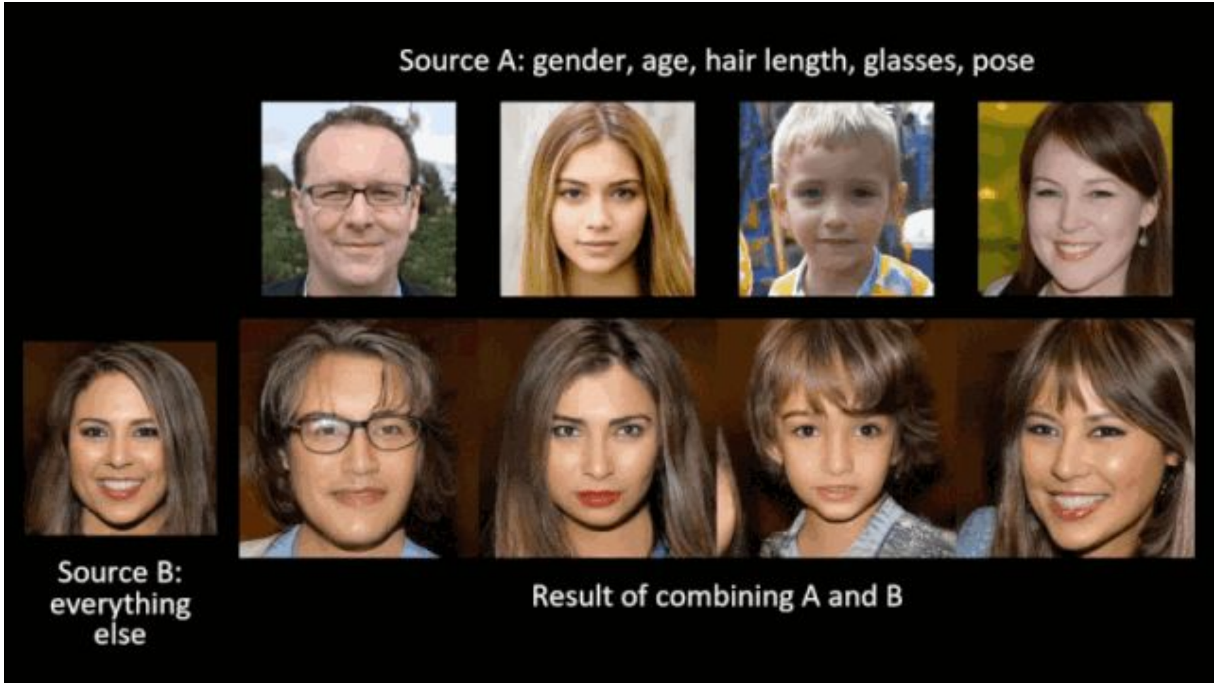
Figure 7: Generated samples

Multiple features of face

Generate face with different features (hair’s color, age, gener,...)

Face combination

Combine source B with each image in source A. More detail here.



Text to image

The input is the text like the sentence and GAN can generate the corresponding image.

Image repainting

Automatically fill missing part for the image

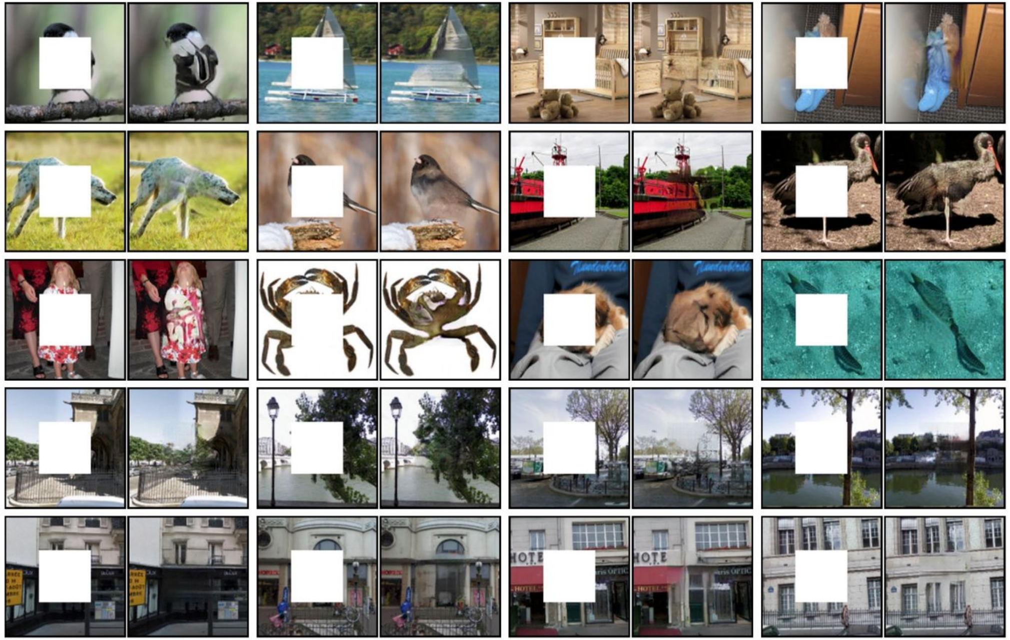
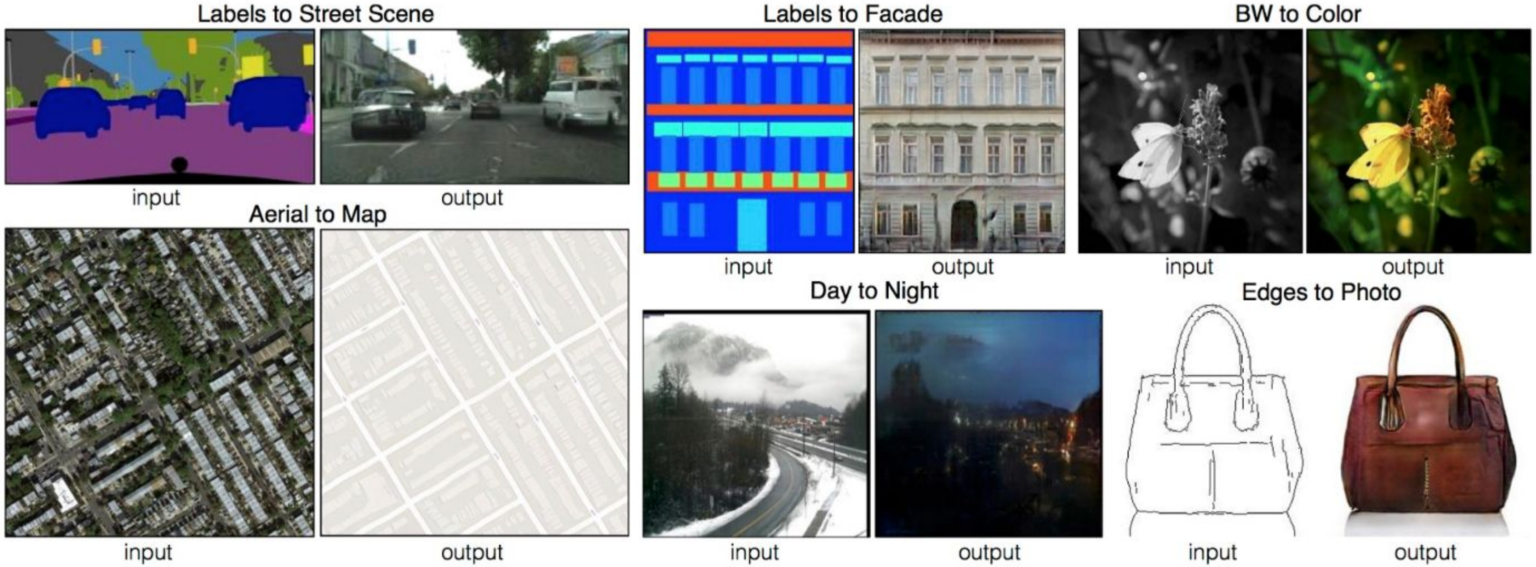


Image to image translation

GAN can learn the mapping between the input and output. It can generate the bag, shoe by the draft only. The interesting demo here.



What we are researching?

- Change the nationality of human's face in picture (like mixed face between japan, taiwan, etc)
- ...